

Assignment 2: Simple Search Engine using Hadoop MapReduce

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Repository: https://github.com/SachaYT1/big_data_2

Section 1: Methodology

1.1 System Architecture

The search engine consists of three main stages:

1. **Data Preparation** — 1000 Wikipedia documents are converted into tab-separated format (`doc_id\ttitle\ttext`) and uploaded to HDFS at `/input/data`
2. **Indexing** — Two Hadoop MapReduce jobs build an inverted index and document statistics, which are then stored in Cassandra
3. **Ranking** — A PySpark application queries the Cassandra index and ranks documents using BM25

```
Wikipedia .txt files (1000 docs)
  → prepare_data.sh → HDFS /input/data (tab-separated)
  → create_index.sh → MapReduce Job 1 (Inverted Index) → HDFS
/indexer/index
                        → MapReduce Job 2 (Doc Stats)      → HDFS
/indexer/doc_stats
  → store_index.sh → Cassandra tables
  → search.sh      → BM25 ranking → Top 10 results
```

1.2 MapReduce Pipeline Design

We use two MapReduce pipelines executed via Hadoop Streaming.

Pipeline 1: Inverted Index

Purpose: For each term, determine which documents contain it, how many times (tf), and in how many documents it appears (df).

Mapper (`mapper1.py`):

- Reads each line: `doc_id\ttitle\ttext`
- Tokenizes text: converts to lowercase, removes non-alphanumeric characters, splits by whitespace
- Emits: `term\tdoc_id` for each token occurrence

Reducer (`reducer1.py`):

- Receives sorted `term\tdoc_id` pairs
- Groups by term, counts occurrences per document → tf (term frequency)

- Counts distinct documents per term → df (document frequency)
- Emits: `term\tdoc_id\ttf\tdf`

Output: HDFS `/indexer/index/`

Pipeline 2: Document Statistics

Purpose: Compute document lengths and corpus-level statistics (N, avgdl) needed for BM25.

Mapper (`mapper2.py`):

- Reads each line: `doc_id\ttitle\text`
- Tokenizes text using the same function as mapper1 (consistency is critical)
- Counts total tokens → doc_length
- Emits: `doc_id\ttitle\tdoc_length`

Reducer (`reducer2.py`):

- Passes through each document's stats
- Accumulates total document count (N) and total word count
- After processing all input, emits a special line: `__CORPUS__\tN\tavgdl`
- **Uses a single reducer** (`-D mapreduce.job.reduces=1`) to ensure global statistics are computed correctly

Output: HDFS `/indexer/doc_stats/`

Tokenization

All components (mapper1, mapper2, query.py) use identical tokenization to ensure consistency:

```
import re
def tokenize(text):
    return re.sub(r'^a-z0-9\s|', ' ', text.lower()).split()
```

This converts text to lowercase, replaces all non-alphanumeric characters with spaces, and splits into tokens.

1.3 Cassandra Database Schema

We designed three tables in the `search_engine` keyspace:

Table 1: `inverted_index`

```
CREATE TABLE inverted_index (
    term text,
    doc_id text,
    tf int,
    df int,
```

```
PRIMARY KEY (term, doc_id)
);
```

Design rationale: Partitioned by `term` with `doc_id` as clustering column. A query `SELECT * FROM inverted_index WHERE term = 'X'` retrieves all postings for a term in a single partition read — exactly what BM25 needs. This is the most performance-critical table since it is queried once per query term.

Table 2: `doc_stats`

```
CREATE TABLE doc_stats (
  doc_id text PRIMARY KEY,
  title text,
  doc_length int
);
```

Design rationale: Partitioned by `doc_id` for O(1) lookup of any document's length and title. Used during BM25 scoring to normalize term frequencies by document length.

Table 3: `corpus_stats`

```
CREATE TABLE corpus_stats (
  id int PRIMARY KEY,
  num_docs int,
  avg_doc_length float
);
```

Design rationale: Single-row table (`id=1`) holding corpus-wide statistics. These values are constant for a given index and are fetched once per query.

1.4 BM25 Implementation

BM25 (Best Matching 25) is a ranking function that scores documents based on query term frequency, document length, and corpus statistics.

Formula:

$$\text{BM25}(q, d) = \sum(t \in q) \log(N / \text{df}(t)) \times ((k_1 + 1) \times \text{tf}(t, d)) / (k_1 \times ((1 - b) + b \times (\text{dl}(d) / \text{avgdl})) + \text{tf}(t, d))$$

Where:

- **N** — total number of documents in the corpus
- **df(t)** — number of documents containing term *t*
- **tf(t,d)** — frequency of term *t* in document *d*

- **dl(d)** — length of document d (in tokens)
- **avgdl** — average document length across the corpus
- **k1 = 1.0** — controls term frequency saturation (higher values give more weight to repeated terms)
- **b = 0.75** — controls document length normalization (1.0 = full normalization, 0.0 = no normalization)

Implementation details (`query.py`):

1. The query string is tokenized using the same function as the mappers
2. For each unique query term, postings are fetched from Cassandra's `inverted_index` table
3. Candidate documents (those containing at least one query term) are collected
4. Document lengths are fetched from `doc_stats`
5. BM25 scores are computed using PySpark RDD API (`sc.parallelize` → `.map` → `.takeOrdered`)
6. Top 10 results are returned sorted by descending score

1.5 Component Interaction

The full pipeline is orchestrated by `app.sh`:

1. `start-services.sh` — starts HDFS, YARN, MapReduce History Server
2. Virtual environment is created and dependencies installed
3. `prepare_data.sh` — uploads documents to HDFS, creates tab-separated format via PySpark
4. `index.sh` calls:
 - `create_index.sh` — runs two Hadoop Streaming MapReduce jobs
 - `store_index.sh` — reads HDFS output via PySpark, loads into Cassandra
5. `search.sh` — executes `query.py` via `spark-submit`

Section 2: Demonstration

2.1 How to Run

Prerequisites:

- Docker and Docker Compose installed
- **Docker Desktop Memory:** at least 6-8 GB (Settings → Resources → Memory)
- **Docker Desktop Disk:** at least 30 GB (Settings → Resources → Disk image size)
- **macOS:** disable AirPlay Receiver to free port 7000 (System Settings → General → AirDrop & Handoff → AirPlay Receiver → OFF)

```
git clone https://github.com/SachaYT1/big_data_2.git
cd big_data_2
docker compose up
```

This starts three containers:

- `cluster-master` — Hadoop/Spark master node (runs the full pipeline)
- `cluster-slave-1` — Hadoop worker node
- `cassandra-server` — Cassandra database

The pipeline runs automatically: data preparation → MapReduce indexing → Cassandra storage → test search query. After completion, the container stays alive for interactive queries:

```
docker exec -it cluster-master bash -c "cd /app && source
.venv/bin/activate && bash search.sh '<your query>'"
```

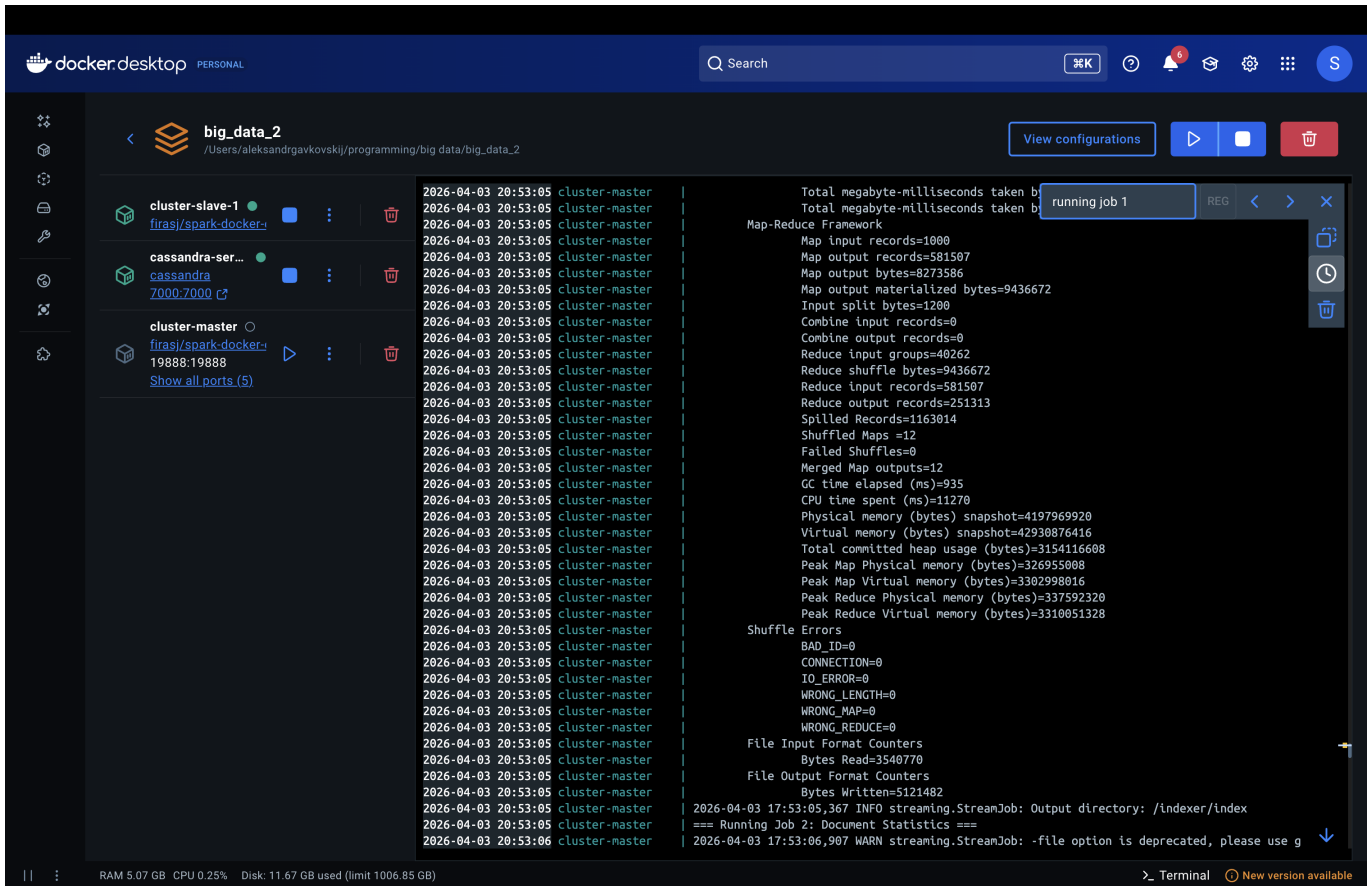
Note: `docker compose down` removes container data (HDFS, Cassandra). A full re-run of `docker compose up` will re-execute the entire pipeline.

2.2 Successful Indexing

MapReduce Job 1 (Inverted Index) started and completed successfully, processing all 1000 documents:

The screenshot shows the Docker Desktop interface with the 'big_data_2' service selected. The logs for the 'cluster-master' container are displayed, showing the execution of the MapReduce job. The logs include the following key messages:

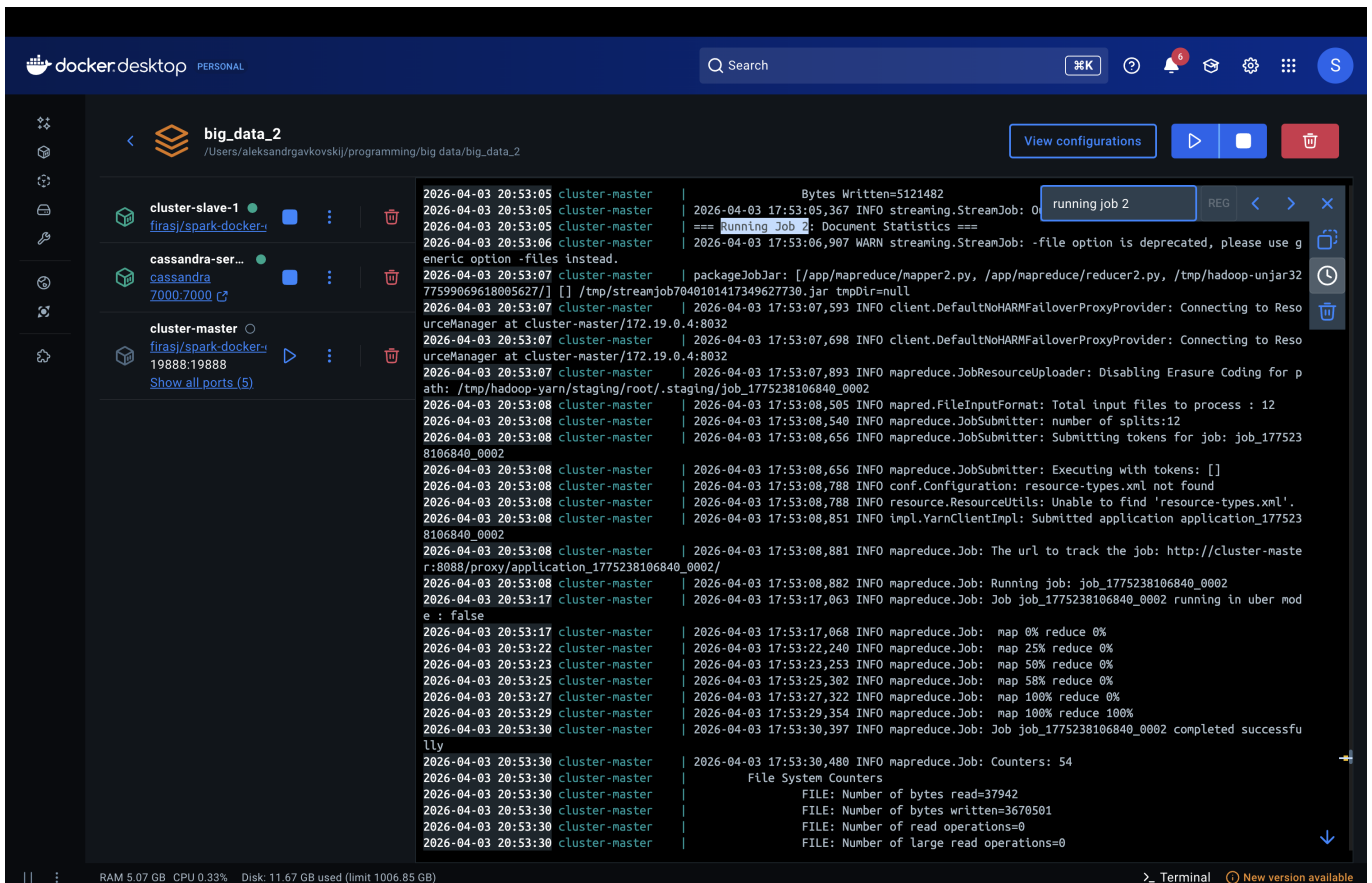
- done data preparation!
- Running full indexing pipeline
- Creating index using MapReduce pipelines
- Input path: /input/data
- Using streaming jar: /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar
- === Running Job 1: Inverted Index ===
- 2026-04-03 17:52:36,906 WARN streaming.StreamJob: -file option is deprecated, please use g
- packageJobJar: [/app/mapreduce/mapper1.py, /app/mapreduce/reducer1.py, /tmp/hadoop-unjar69
- 32727878617329558/] [] /tmp/streamjob4391155160510267945.jar tmpDir=null
- 2026-04-03 17:52:37,562 INFO client.DefaultNoHARMAFollowerProxyProvider: Connecting to Reso
- urceManager at cluster-master/172.19.0.4:8032
- 2026-04-03 17:52:37,668 INFO client.DefaultNoHARMAFollowerProxyProvider: Connecting to Reso
- urceManager at cluster-master/172.19.0.4:8032
- 2026-04-03 17:52:37,887 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for p
- ath: /tmp/hadoop-yarn/staging/root/.staging/job_1775238106840_0001
- 2026-04-03 17:52:39,425 INFO mapred.FileInputFormat: Total input files to process : 12
- 2026-04-03 17:52:40,323 INFO mapreduce.JobSubmitter: number of splits:12
- 2026-04-03 17:52:40,473 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_177523
- 8106840_0001
- 2026-04-03 17:52:40,474 INFO mapreduce.JobSubmitter: Executing with tokens: []
- 2026-04-03 17:52:40,598 INFO conf.Configuration: resource-types.xml not found
- 2026-04-03 17:52:40,599 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
- 2026-04-03 17:52:40,771 INFO impl.YarnClientImpl: Submitted application application_177523
- 8106840_0001
- 2026-04-03 17:52:40,825 INFO mapreduce.Job: The url to track the job: http://cluster-naste
- r:8088/proxy/application_1775238106840_0001/
- 2026-04-03 17:52:40,826 INFO mapreduce.Job: Running job: job_1775238106840_0001
- 2026-04-03 17:52:48,005 INFO mapreduce.Job: Job job_1775238106840_0001 running in uber mod
- e : false
- 2026-04-03 17:52:48,010 INFO mapreduce.Job: map 0% reduce 0%
- 2026-04-03 17:52:54,118 INFO mapreduce.Job: map 50% reduce 0%
- 2026-04-03 17:52:59,186 INFO mapreduce.Job: map 100% reduce 0%
- 2026-04-03 17:53:04,238 INFO mapreduce.Job: map 100% reduce 100%
- 2026-04-03 17:53:05,286 INFO mapreduce.Job: Job job_1775238106840_0001 completed successfu
- lly
- 2026-04-03 17:53:05,367 INFO mapreduce.Job: Counters: 55
- File System Counters
- FILE: Number of bytes read=9436606
- FILE: Number of bytes written=22467790
- FILE: Number of read operations=0



The screenshot shows the Docker Desktop interface with the 'big_data_2' container selected. The logs for the 'cluster-master' service are displayed, showing the progress of a MapReduce job. The logs indicate that the job is running on a Spark cluster and is currently in the 'Map-Reduce Framework' phase. The logs show the number of records processed, the time taken, and the progress of the job. The logs also show the progress of the job, including the number of records processed and the time taken.

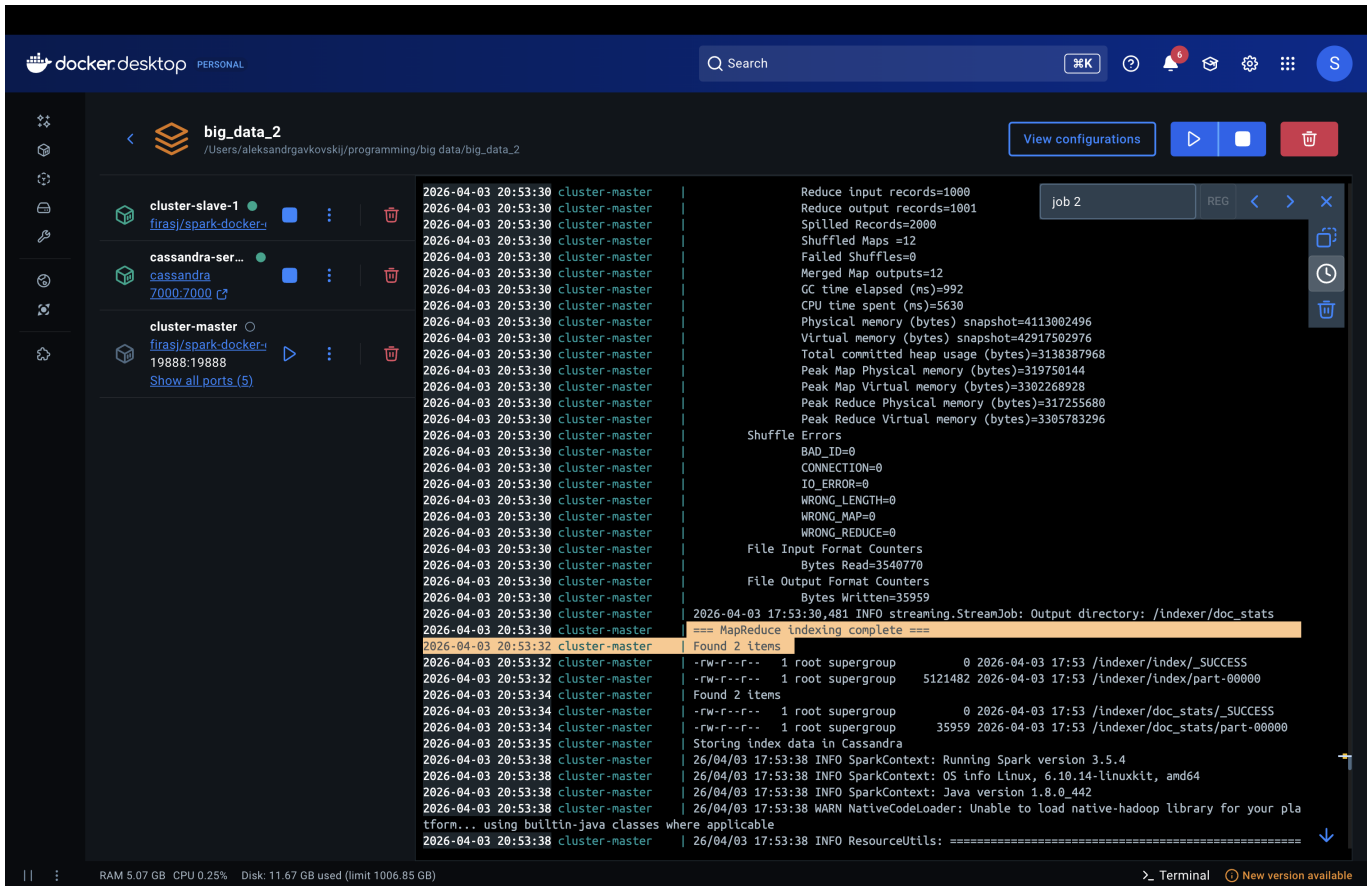
```
2026-04-03 20:53:05 cluster-master | Total megabyte-milliseconds taken b
2026-04-03 20:53:05 cluster-master | Total megabyte-milliseconds taken b
2026-04-03 20:53:05 cluster-master | Map-Reduce Framework
2026-04-03 20:53:05 cluster-master | Map input records=1000
2026-04-03 20:53:05 cluster-master | Map output records=581507
2026-04-03 20:53:05 cluster-master | Map output bytes=8273586
2026-04-03 20:53:05 cluster-master | Map output materialized bytes=9436672
2026-04-03 20:53:05 cluster-master | Input split bytes=1200
2026-04-03 20:53:05 cluster-master | Combine input records=0
2026-04-03 20:53:05 cluster-master | Combine output records=0
2026-04-03 20:53:05 cluster-master | Reduce input groups=40262
2026-04-03 20:53:05 cluster-master | Reduce shuffle bytes=9436672
2026-04-03 20:53:05 cluster-master | Reduce input records=581507
2026-04-03 20:53:05 cluster-master | Reduce output records=251313
2026-04-03 20:53:05 cluster-master | Spilled Records=1163014
2026-04-03 20:53:05 cluster-master | Shuffled Maps =12
2026-04-03 20:53:05 cluster-master | Failed Shuffles=0
2026-04-03 20:53:05 cluster-master | Merged Map outputs=12
2026-04-03 20:53:05 cluster-master | GC time elapsed (ms)=935
2026-04-03 20:53:05 cluster-master | CPU time spent (ms)=11270
2026-04-03 20:53:05 cluster-master | Physical memory (bytes) snapshot=4197969920
2026-04-03 20:53:05 cluster-master | Virtual memory (bytes) snapshot=42930876416
2026-04-03 20:53:05 cluster-master | Total committed heap usage (bytes)=3154116608
2026-04-03 20:53:05 cluster-master | Peak Map Physical memory (bytes)=326955008
2026-04-03 20:53:05 cluster-master | Peak Map Virtual memory (bytes)=3302998016
2026-04-03 20:53:05 cluster-master | Peak Reduce Physical memory (bytes)=337592320
2026-04-03 20:53:05 cluster-master | Peak Reduce Virtual memory (bytes)=3310851328
2026-04-03 20:53:05 cluster-master | Shuffle Errors
2026-04-03 20:53:05 cluster-master | BAD_ID=0
2026-04-03 20:53:05 cluster-master | CONNECTION=0
2026-04-03 20:53:05 cluster-master | IO_ERROR=0
2026-04-03 20:53:05 cluster-master | WRONG_LENGTH=0
2026-04-03 20:53:05 cluster-master | WRONG_MAP=0
2026-04-03 20:53:05 cluster-master | WRONG_REDUCE=0
2026-04-03 20:53:05 cluster-master | File Input Format Counters
2026-04-03 20:53:05 cluster-master | Bytes Read=3540770
2026-04-03 20:53:05 cluster-master | File Output Format Counters
2026-04-03 20:53:05 cluster-master | Bytes Written=5121482
2026-04-03 17:53:05,367 INFO streaming.StreamJob: Output directory: /Indexer/Index
2026-04-03 17:53:06,907 WARN streaming.StreamJob: -file option is deprecated, please use g
```

MapReduce Job 2 (Document Statistics) completed with a single reducer, computing corpus-level statistics:



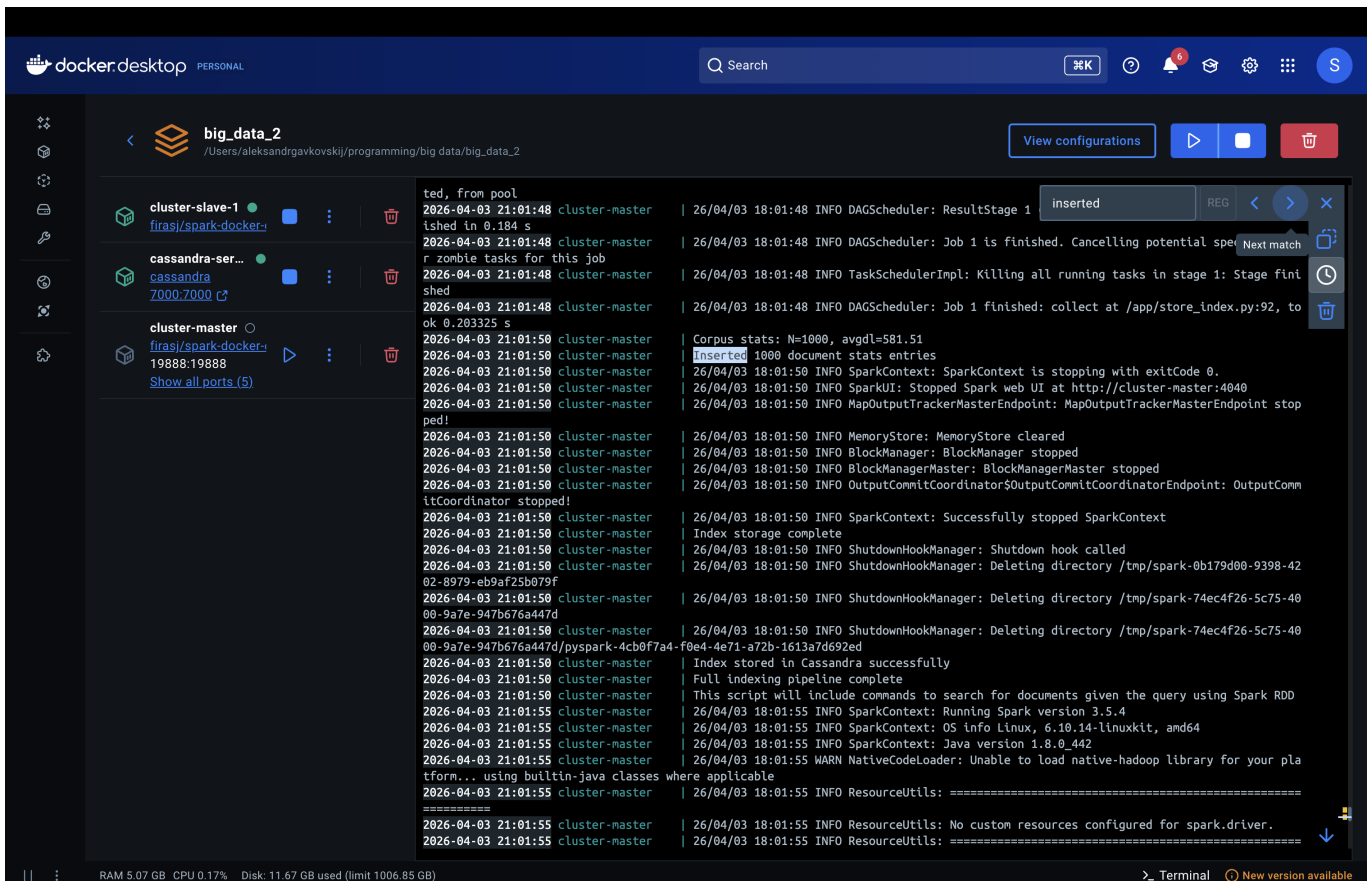
The screenshot shows the Docker Desktop interface with the 'big_data_2' container selected. The logs for the 'cluster-master' service are displayed, showing the progress of a MapReduce job. The logs indicate that the job is running on a Spark cluster and is currently in the 'Map-Reduce Framework' phase. The logs show the number of records processed, the time taken, and the progress of the job. The logs also show the progress of the job, including the number of records processed and the time taken.

```
2026-04-03 20:53:05 cluster-master | Bytes Written=5121482
2026-04-03 17:53:05,367 INFO streaming.StreamJob: O
2026-04-03 17:53:06,907 WARN streaming.StreamJob: -file option is deprecated, please use g
2026-04-03 20:53:06 cluster-master | packageJobJar: [/app/mapreduce/mapper2.py, /app/mapreduce/reducer2.py, /tmp/hadoop-unjar32
2026-04-03 20:53:07 cluster-master | 7759969618005627/] [] /tmp/streamjob7040101417349627730.jar tmpDir=null
2026-04-03 20:53:07 cluster-master | 2026-04-03 17:53:07,593 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to Reso
2026-04-03 20:53:07 cluster-master | 2026-04-03 17:53:07,698 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to Reso
2026-04-03 20:53:07 cluster-master | 2026-04-03 17:53:07,893 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for p
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,505 INFO mapred.FileInputFormat: Total input files to process : 12
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,540 INFO mapreduce.JobSubmitter: number of splits:12
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,656 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_177523
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,656 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,788 INFO conf.Configuration: resource-types.xml not found
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,788 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,851 INFO impl.YarnClientImpl: Submitted application application_177523
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,881 INFO mapreduce.Job: The url to track the job: http://cluster-maste
2026-04-03 20:53:08 cluster-master | 2026-04-03 17:53:08,882 INFO mapreduce.Job: Running job: job_1775238106840_0002
2026-04-03 20:53:17 cluster-master | 2026-04-03 17:53:17,063 INFO mapreduce.Job: Job job_1775238106840_0002 running in uber mod
2026-04-03 20:53:17 cluster-master | 2026-04-03 17:53:17,068 INFO mapreduce.Job: map 0% reduce 0%
2026-04-03 20:53:22 cluster-master | 2026-04-03 17:53:22,240 INFO mapreduce.Job: map 25% reduce 0%
2026-04-03 20:53:23 cluster-master | 2026-04-03 17:53:23,253 INFO mapreduce.Job: map 58% reduce 0%
2026-04-03 20:53:25 cluster-master | 2026-04-03 17:53:25,302 INFO mapreduce.Job: map 58% reduce 0%
2026-04-03 20:53:27 cluster-master | 2026-04-03 17:53:27,322 INFO mapreduce.Job: map 100% reduce 0%
2026-04-03 20:53:29 cluster-master | 2026-04-03 17:53:29,354 INFO mapreduce.Job: map 100% reduce 100%
2026-04-03 20:53:30 cluster-master | 2026-04-03 17:53:30,397 INFO mapreduce.Job: Job job_1775238106840_0002 completed successfu
2026-04-03 20:53:30 cluster-master | 2026-04-03 17:53:30,480 INFO mapreduce.Job: Counters: 54
2026-04-03 20:53:30 cluster-master | File System Counters
2026-04-03 20:53:30 cluster-master | FILE: Number of bytes read=37942
2026-04-03 20:53:30 cluster-master | FILE: Number of bytes written=3670501
2026-04-03 20:53:30 cluster-master | FILE: Number of read operations=0
2026-04-03 20:53:30 cluster-master | FILE: Number of large read operations=0
```

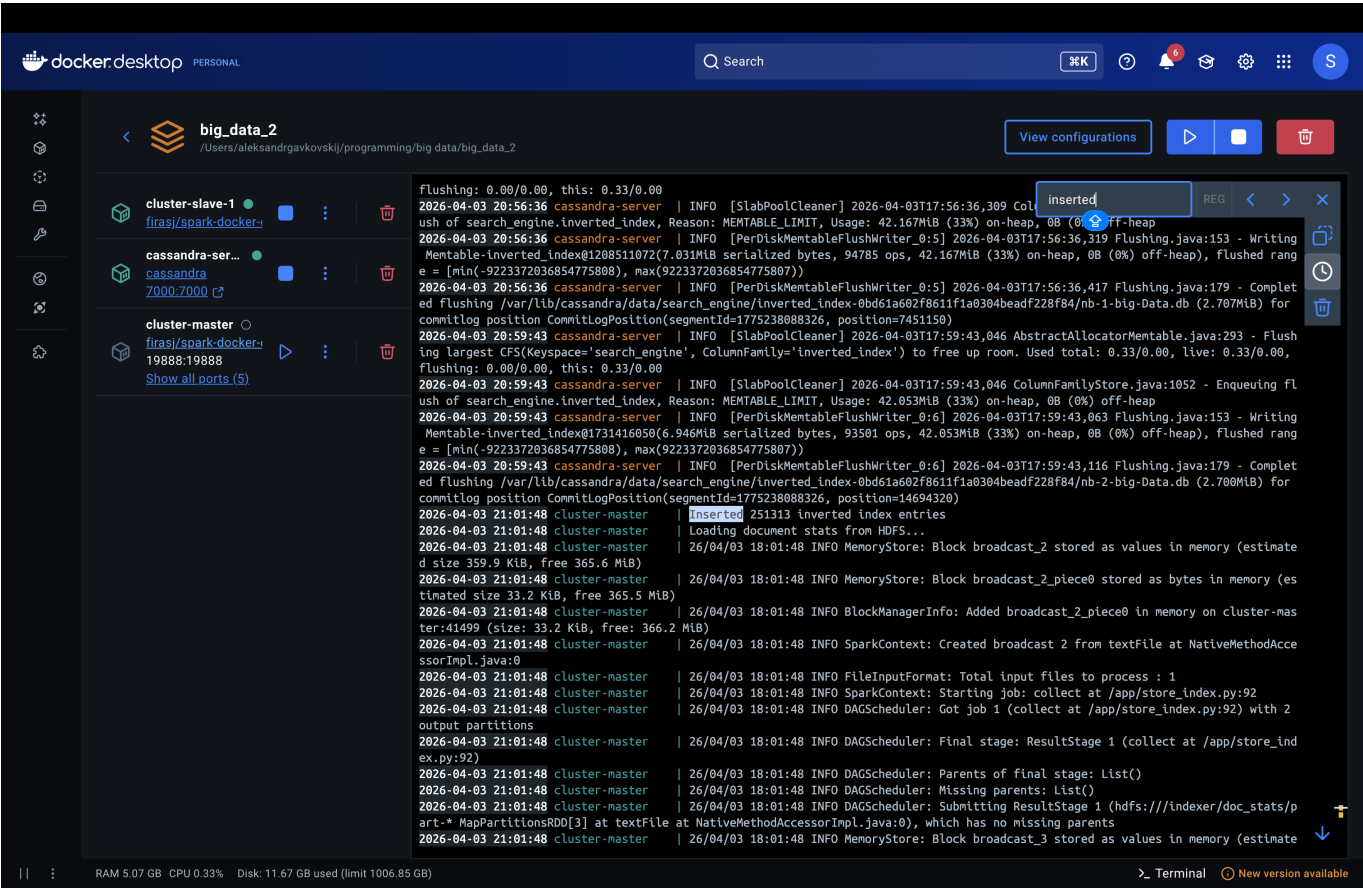



```
2026-04-03 20:53:30 cluster-master Reduce input records=1000
2026-04-03 20:53:30 cluster-master Reduce output records=1001
2026-04-03 20:53:30 cluster-master Spilled Records=2000
2026-04-03 20:53:30 cluster-master Shuffled Maps =12
2026-04-03 20:53:30 cluster-master Failed Shuffles=0
2026-04-03 20:53:30 cluster-master Merged Map outputs=12
2026-04-03 20:53:30 cluster-master GC time elapsed (ms)=992
2026-04-03 20:53:30 cluster-master CPU time spent (ms)=5630
2026-04-03 20:53:30 cluster-master Physical memory (bytes) snapshot=4113002496
2026-04-03 20:53:30 cluster-master Virtual memory (bytes) snapshot=42917502976
2026-04-03 20:53:30 cluster-master Total committed heap usage (bytes)=3138387968
2026-04-03 20:53:30 cluster-master Peak Map Physical memory (bytes)=319750144
2026-04-03 20:53:30 cluster-master Peak Map Virtual memory (bytes)=3302268928
2026-04-03 20:53:30 cluster-master Peak Reduce Physical memory (bytes)=317255680
2026-04-03 20:53:30 cluster-master Peak Reduce Virtual memory (bytes)=3305783296
2026-04-03 20:53:30 cluster-master Shuffle Errors
2026-04-03 20:53:30 cluster-master BAD_ID=0
2026-04-03 20:53:30 cluster-master CONNECTION=0
2026-04-03 20:53:30 cluster-master IO_ERROR=0
2026-04-03 20:53:30 cluster-master WRONG_LENGTH=0
2026-04-03 20:53:30 cluster-master WRONG_MAP=0
2026-04-03 20:53:30 cluster-master WRONG_REDUCE=0
2026-04-03 20:53:30 cluster-master File Input Format Counters
2026-04-03 20:53:30 cluster-master Bytes Read=3540770
2026-04-03 20:53:30 cluster-master File Output Format Counters
2026-04-03 20:53:30 cluster-master Bytes Written=35959
2026-04-03 17:53:30,481 INFO streaming.StreamJob: Output directory: /indexer/doc_stats
2026-04-03 20:53:32 cluster-master == MapReduce indexing complete ==
2026-04-03 20:53:32 cluster-master Found 2 items
2026-04-03 20:53:32 cluster-master -rw-r--r-- 1 root supergroup 0 2026-04-03 17:53 /indexer/index/_SUCCESS
2026-04-03 20:53:32 cluster-master -rw-r--r-- 1 root supergroup 5121482 2026-04-03 17:53 /indexer/index/part-00000
2026-04-03 20:53:34 cluster-master Found 2 items
2026-04-03 20:53:34 cluster-master -rw-r--r-- 1 root supergroup 0 2026-04-03 17:53 /indexer/doc_stats/_SUCCESS
2026-04-03 20:53:34 cluster-master -rw-r--r-- 1 root supergroup 35959 2026-04-03 17:53 /indexer/doc_stats/part-00000
2026-04-03 20:53:35 cluster-master Storing index data in Cassandra
2026-04-03 17:53:38 INFO SparkContext: Running Spark version 3.5.4
2026-04-03 17:53:38 INFO SparkContext: OS info Linux, 6.10.14-linuxkit, amd64
2026-04-03 17:53:38 INFO SparkContext: Java version 1.8.0_442
2026-04-03 17:53:38 WARN NativeCodeLoader: Unable to load native-hadoop library for your pla
tform... using builtin-java classes where applicable
2026-04-03 17:53:38 INFO ResourceUtils: =====
```

Cassandra loading — the index was successfully stored in all three tables. A total of 251,313 inverted index entries and 1000 document stats entries were inserted:

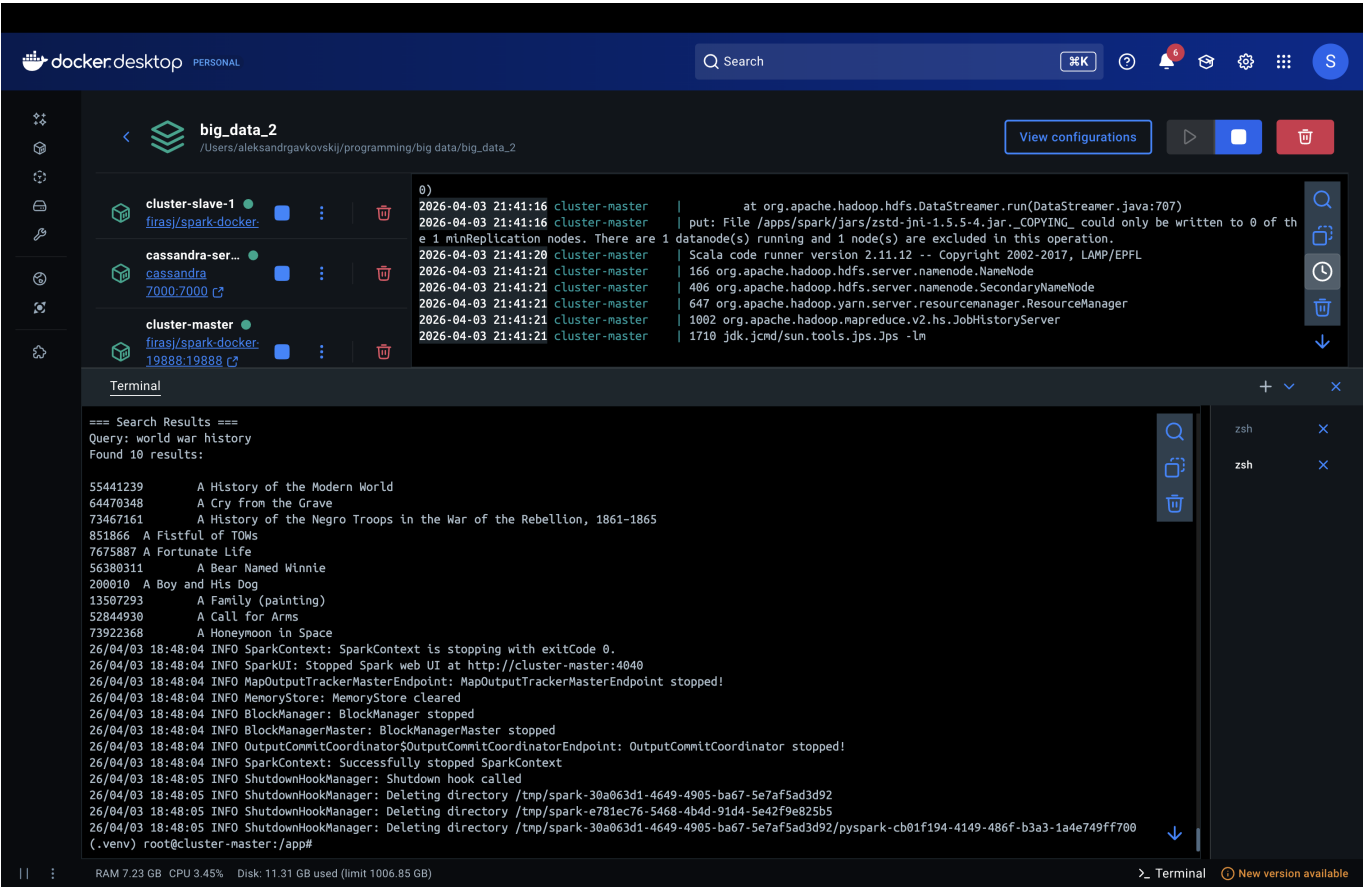


```
ted, from pool
2026-04-03 21:01:48 cluster-master ished in 0.184 s
2026-04-03 21:01:48 cluster-master r zombie tasks for this job
2026-04-03 21:01:48 cluster-master shed
2026-04-03 21:01:48 cluster-master ok 0.203325 s
2026-04-03 21:01:50 cluster-master Corpus stats: N=1000, avgdl=581.51
2026-04-03 21:01:50 cluster-master Inserted 1000 document stats entries
2026-04-03 18:01:50 INFO SparkContext: SparkContext is stopping with exitCode 0.
2026-04-03 18:01:50 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
2026-04-03 18:01:50 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stop
ped!
2026-04-03 18:01:50 INFO MemoryStore: MemoryStore cleared
2026-04-03 18:01:50 INFO BlockManager: BlockManager stopped
2026-04-03 18:01:50 INFO BlockManagerMaster: BlockManagerMaster stopped
2026-04-03 18:01:50 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputComm
itCoordinator stopped!
2026-04-03 18:01:50 INFO SparkContext: Successfully stopped SparkContext
2026-04-03 18:01:50 INFO Index storage complete
2026-04-03 18:01:55 INFO ShutdownHookManager: Shutdown hook called
2026-04-03 18:01:55 INFO ShutdownHookManager: Deleting directory /tmp/spark-0b179d00-9398-42
02-8979-eb9af25b079f
2026-04-03 18:01:55 INFO ShutdownHookManager: Deleting directory /tmp/spark-74ec4f26-5c75-40
00-9a7e-947b676a447d
2026-04-03 18:01:55 INFO ShutdownHookManager: Deleting directory /tmp/spark-74ec4f26-5c75-40
00-9a7e-947b676a447d/pyspark-4cb0f7a4-f0e4-4e71-a72b-1613a7d692ed
2026-04-03 18:01:55 cluster-master Index stored in Cassandra successfully
2026-04-03 18:01:55 cluster-master Full indexing pipeline complete
2026-04-03 18:01:55 cluster-master This script will include commands to search for documents given the query using Spark RDD
2026-04-03 18:01:55 INFO SparkContext: Running Spark version 3.5.4
2026-04-03 18:01:55 INFO SparkContext: OS info Linux, 6.10.14-linuxkit, amd64
2026-04-03 18:01:55 INFO SparkContext: Java version 1.8.0_442
2026-04-03 18:01:55 cluster-master tform... using builtin-java classes where applicable
2026-04-03 18:01:55 INFO ResourceUtils: =====
2026-04-03 18:01:55 cluster-master 2026-04-03 18:01:55 INFO ResourceUtils: No custom resources configured for spark.driver.
2026-04-03 18:01:55 cluster-master 2026-04-03 18:01:55 INFO ResourceUtils: =====
```



2.3 Search Queries and Results

Query 1: "world war history"



Analysis: The top results are highly relevant. "A History of the Modern World" and "A History of the Negro Troops in the War of the Rebellion" rank highest because they contain all three query terms ("world", "war", "history") with high term frequencies. "A Cry from the Grave" — a documentary about the Srebrenica massacre — ranks second due to frequent mentions of "war" and "history". Documents like "A Fistful of TOWs" (a war game) and "A Call for Arms" appear lower because they match fewer query terms but still have high tf for "war".

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big_data_2

/Users/aleksandr.gavkovskij/programming/big_data/big_data_2

View configurations

Name	Status	Ports	Actions
cluster-slave-1 firasj/spark-docker	Running		[Stop] [Refresh] [Delete]
cassandra-ser... cassandra 7000:7000	Running		[Stop] [Refresh] [Delete]
cluster-master firasj/spark-docker 19888:19888	Running		[Stop] [Refresh] [Delete]

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0)
2026-04-03 21:41:16 cluster-master | at org.apache.hadoop.hdfs.DataStreamer.run(DataStreamer.java:707)
2026-04-03 21:41:16 cluster-master | put: File /apps/spark/jars/zstd-jni-1.5.5-4.jar._COPYING_ could only be written to 0 of th
e 1 minReplication nodes. There are 1 datanode(s) running and 1 node(s) are excluded in this operation.
2026-04-03 21:41:20 cluster-master | Scala code runner version 2.11.12 -- Copyright 2002-2017, LAMP/EPFL
2026-04-03 21:41:21 cluster-master | 166 org.apache.hadoop.hdfs.server.namenode.NameNode
2026-04-03 21:41:21 cluster-master | 406 org.apache.hadoop.hdfs.server.namenode.SecondaryNameNode
2026-04-03 21:41:21 cluster-master | 647 org.apache.hadoop.yarn.resourcemanager.ResourceManager
2026-04-03 21:41:21 cluster-master | 1002 org.apache.hadoop.mapreduce.v2.Hs.JobHistoryServer
2026-04-03 21:41:21 cluster-master | 1710 jdk.jcmd/sun.tools.jpss.Jps -lm
    
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=== Search Results ===
Query: music album rap
Found 10 results:

31452302      A Cry Farewell
67333877     A Gangsta's Pain
673878      A Grand Don't Come for Free
39854476     A Joyful Noise Unto The Creator
55178618     A Boogie wit da Hoodie discography
16968794     A Little Bit Longer
23619277     A Brand You Can Trust
3274533      A Flight and a Crash
70243173     A Concert in Berlin
29045544     A Christmas Cornucopia

26/04/03 18:48:55 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/04/03 18:48:55 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
26/04/03 18:48:55 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/04/03 18:48:55 INFO MemoryStore: MemoryStore cleared
26/04/03 18:48:55 INFO BlockManager: BlockManager stopped
26/04/03 18:48:55 INFO BlockManagerMaster: BlockManagerMaster stopped
26/04/03 18:48:55 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/04/03 18:48:55 INFO SparkContext: Successfully stopped SparkContext
26/04/03 18:48:56 INFO ShutdownHookManager: Shutdown hook called
26/04/03 18:48:56 INFO ShutdownHookManager: Deleting directory /tmp/spark-a4e88ad8-0a6c-4656-8eb7-6c0fb56725e9
26/04/03 18:48:56 INFO ShutdownHookManager: Deleting directory /tmp/spark-a180b9cd-1ad0-4fae-b670-b4b8012c53e2/pyspark-a3a7ef42-ff4e-453f-8029-a95e46cd359d
26/04/03 18:48:56 INFO ShutdownHookManager: Deleting directory /tmp/spark-a180b9cd-1ad0-4fae-b670-b4b8012c53e2
(.venv) root@cluster-master:/app#
    
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RAM 6.50 GB CPU 0.42% Disk: 11.31 GB used (limit 1006.85 GB)

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55178618	A Boogie wit da Hoodie discography
16968794	A Little Bit Longer
23619277	A Brand You Can Trust
3274533	A Flight and a Crash
70243173	A Concert in Berlin
29045544	A Christmas Cornucopia

Query 3: "death metal band"

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big_data_2
 /Users/aleksandravkovskij/programming/big data/big_data_2

View configurations

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Container Name	Status	IP Address	Ports	Actions	Logs
cluster-slave-1 firasj/spark-docker:	Running	19888	2181, 7070		<pre> 2026-04-03 21:41:16 cluster-master at org.apache.hadoop.hdfs.DataStreamer.run(DataStreamer.java:707) 2026-04-03 21:41:16 cluster-master put: File /apps/spark/jars/zstd-jni-1.5.5-4.jar._COPYING_ could only be written to 0 of th e 1 minReplication nodes. There are 1 node(s) are excluded in this operation. 2026-04-03 21:41:20 cluster-master Scala code runner version 2.11.12 -- Copyright 2002-2017, LAMP/EPFL 2026-04-03 21:41:21 cluster-master 166 org.apache.hadoop.hdfs.server.namenode.NameNode 2026-04-03 21:41:21 cluster-master 406 org.apache.hadoop.hdfs.server.namenode.SecondaryNameNode 2026-04-03 21:41:21 cluster-master 647 org.apache.hadoop.yarn.server.resourcemanager.ResourceManager 2026-04-03 21:41:21 cluster-master 1002 org.apache.hadoop.mapreduce.v2.hs.JobHistoryServer 2026-04-03 21:41:21 cluster-master 1710 jdk.jcmd/sun.tools.Jps -lm </pre>
cassandra-ser... cassandra 7000-7000	Running	7000			
cluster-master firasj/spark-docker: 19888.19888	Running	19888	2181, 7070		

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26/04/03 18:46:53 INFO DAGScheduler: Job 0 finished: takeOrdered at /app/query.py:106, took 6.646988 s

=== Search Results ===

Query: death metal band

Found 10 results:

- 2054290 A Blaze in the Northern Sky
- 46923580 A Hill to Die Upon
- 10031136 A Decade in the Grave
- 2873166 A Celebration of Gullt
- 55638628 A Decade of Destruction
- 31452302 A Cry Farewell
- 2983588 A Collection of Metal
- 61899060 A Dawn to Fear
- 6597753 A II Z
- 9413554 A Haunting Curse

26/04/03 18:46:53 INFO SparkContext: SparkContext is stopping with exitCode 0.

26/04/03 18:46:53 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040

26/04/03 18:46:53 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!

26/04/03 18:46:53 INFO MemoryStore: MemoryStore cleared

26/04/03 18:46:53 INFO BlockManager: BlockManager stopped

26/04/03 18:46:53 INFO BlockManagerMaster: BlockManagerMaster stopped

26/04/03 18:46:53 INFO OutputCommitCoordinator\$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!

26/04/03 18:46:53 INFO SparkContext: Successfully stopped SparkContext

26/04/03 18:46:54 INFO ShutdownHookManager: Shutdown hook called

26/04/03 18:46:54 INFO ShutdownHookManager: Deleting directory /tmp/spark-d3c9269e-5a2a-4ae5-a8e1-78f92b66564a

26/04/03 18:46:54 INFO ShutdownHookManager: Deleting directory /tmp/spark-d3c9269e-5a2a-4ae5-a8e1-78f92b66564a/pyspark-519b5afb-fd6f-4275-b6c9-477facfd34c

RAM 6.39 GB CPU 1.77% Disk: 11.31 GB used (limit 1006.85 GB)

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2054290	A Blaze in the Northern Sky
46923580	A Hill to Die Upon
10031136	A Decade in the Grave
2873166	A Celebration of Guilt
55638628	A Decade of Destruction
31452302	A Cry Farewell
2983588	A Collection of Metal
61899060	A Dawn to Fear
6597753	A II Z
9413554	A Haunting Curse

Analysis: The results are very relevant to the query. "A Blaze in the Northern Sky" (Darkthrone's iconic black/death metal album) ranks first due to high combined tf for all three terms. "A Hill to Die Upon" and "A Decade in the Grave" (Six Feet Under box set) follow closely — both are articles about death metal bands with frequent mentions of "death", "metal", and "band". "A Celebration of Guilt" (Arsis album) and "A Collection of Metal" also rank high due to strong term overlap. The BM25 document length normalization ($b=0.75$) helps shorter, focused articles about specific bands rank above longer, more general articles.

2.4 Reflections

Overall, the search results were relevant for all three queries. For "death metal band", the top results were actual death metal albums and compilations, which makes sense. For "music album rap", we got rap albums and discographies. For "world war history", most results were clearly related to wars and history.

BM25 handles multi-word queries well because it sums up scores for each query term separately. So a document that matches all three words gets a much higher score than one matching just one. Also, rare words like "rap" or "death" contribute more to the score than common words like "music" — this is the IDF part of the formula doing its job.

One thing I noticed is that some results for "world war history" (like "A Bear Named Winnie" or "A Boy and His Dog") seem unrelated at first, but they actually do mention wars in their text. This shows that BM25 is a statistical method — it counts words, it does not understand meaning. If a document mentions "war" enough times, it will rank high regardless of what the article is actually about.

Document length normalization ($b=0.75$) also had a visible effect — shorter focused articles ranked higher than long general ones, because BM25 adjusts for the fact that longer documents naturally contain more word occurrences.